

Package ‘ButterflyLifespan’

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Type Package

Title Butterfly Longevity Estimation

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Description Provides methods and data to enable reproduction of the results presented in ``Estimation of adult butterfly longevity using long-term citizen-science count data" (Clarke, Dennis and McCrea 2026)

License GPL (>= 2)

Depends R(>= 3.5),dplyr,reshape2

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NeedsCompilation no

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analysis	<i>Analysis code for fitting a GAI stopover model to a single year of data.</i>
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Description

Code used for fitting a single year GAI stopover model. The phi parameter is used to derive lifespan estimates. Data can be added either in a "daily" or "weekly" format.

Usage

```
analysis(data, year, setup_level)
```

Arguments

data	dataframe containing the data to analysis
year	character value corresponding to the the year to analyse from the data
setup_level	character value "daily" or "weekly" dependent on how the data was sampled.

Value

A list object containing three lists. The first containing the output used for deriving lifespan estimates and the second containing each output from each random start. The third states the likelihood for each random start.

Examples

```
library(ButterflyLifespan)

# lazy load data
data(dark_green_fritillary_weekly)
# year to analyse
year <- "2018"
# weekly sampling
setup_level <- "weekly"

# run analysis
results <- analysis(dark_green_fritillary_weekly, year, setup_level)
```

analysis_multiyear	<i>Analysis code for fitting a multiyear GAI stopover model to data spanning multiple years.</i>
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Description

Code used for fitting a multiyear GAI stopover model to data spanning multiple years. The phi parameter is used to derive lifespan estimates and can be modelled to be "fixed" across all years, to be fully "variable" year-to-year or modelled to change gradually with a "slope" term. Data can be added either in a "daily" or "weekly" format.

Usage

```
analysis_multiyear(data, setup_level, phi_type)
```

Arguments

data	dataframe containing the data to analysis
setup_level	character value "daily" or "weekly" dependent on how the data was sampled.
phi_type	character value "doubleslope", "slope", "variable" or "fixed"

Value

A list object containing three lists. The first containing the output used for deriving lifespan estimates and the second containing each output from each random start. The third states the likelihood for each random start.

Examples

```
library(ButterflyLifespan)

# lazy load data
data(orange_tip_weekly)
# year to analyse
year <- "2018"
# weekly sampling
setup_level <- "weekly"
# phi type
phi_type = "slope"

# run analysis
results <- analysis_multiyear(orange_tip_weekly, setup_level, phi_type)
```

```
dark_green_fritillary_daily
```

Dark Green Fritillary

Description

Daily site, year, and count data for dark green fritillary

Usage

```
data(dark_green_fritillary_daily )
```

Format

A dataframe with 5 columns and 581129 rows. The columns are DAYNO (day number), WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), COUNT (count), and DATE (date).

Examples

```
library(ButterflyLifespan)

data(dark_green_fritillary_daily )
head(dark_green_fritillary_daily )
plot(dark_green_fritillary_daily $COUNT)
```

```
dark_green_fritillary_weekly
```

Dark Green Fritillary Weekly

Description

Weekly site, year, and count data for dark green fritillary

Usage

```
data(dark_green_fritillary_weekly)
```

Format

A dataframe with 5 columns and 82420 rows. The columns are WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), and COUNT (count).

Examples

```
library(ButterflyLifespan)

data(dark_green_fritillary_weekly)
head(dark_green_fritillary_weekly)
plot(dark_green_fritillary_weekly$COUNT)
```

marbled_white_daily	<i>Marbled White</i>
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Description

Daily site, year, and count data for marbled white

Usage

```
data(marbled_white_daily)
```

Format

A dataframe with 5 columns and 1048575 rows. The columns are DAYNO (day number), WEE-KNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), COUNT (count), and DATE (date).

Examples

```
library(ButterflyLifespan)

data(marbled_white_daily)
head(marbled_white_daily)
plot(marbled_white_daily$COUNT)
```

marbled_white_weekly	<i>Marbled White Weekly</i>
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Description

Weekly site, year, and count data for marbled white

Usage

```
data(marbled_white_weekly)
```

Format

A dataframe with 5 columns and 206778 rows. The columns are WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), and COUNT (count).

Examples

```
library(ButterflyLifespan)

data(marbled_white_weekly)
head(marbled_white_weekly)
plot(marbled_white_weekly$COUNT)
```

orange_tip_daily	<i>Orange Tip Daily</i>
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Description

Daily site, year, and count data for orange tip

Usage

```
data(orange_tip_daily)
```

Format

A dataframe with 5 columns and 1048575 rows. The columns are DAYNO (day number), WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), COUNT (count), and DATE (date).

Examples

```
library(ButterflyLifespan)

data(orange_tip_daily)
head(orange_tip_daily)
plot(orange_tip_daily$COUNT)
```

orange_tip_weekly	<i>Orange Tip Weekly</i>
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Description

Weekly site, year, and count data for orange tip

Usage

```
data(orange_tip_weekly)
```

Format

A dataframe with 5 columns and 346164 rows. The columns are WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), and COUNT (count).

Examples

```
library(ButterflyLifespan)

data(orange_tip_weekly)
head(orange_tip_weekly)
plot(orange_tip_weekly$COUNT)
```

simulation	<i>Simulation code to assess model performance.</i>
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Description

Simulation code to be used to assess biases in model output. Associated with number of days sampled per week or number of sites sampled.

Usage

```
simulation(
  setup_level,
  env_cov.type,
  a.type,
  dist.type,
  B,
  cov_sim_type,
  n.iter,
  dpw,
  meth,
```

```

    convert_weekly,
    vary_phi,
    NA_type,
    nS,
    year
  )

```

Arguments

setup_level	character value "daily" or "weekly" dependent on how the data is to be sampled.
env_cov.type	how to add environmental covariate data, either "lin", "quad" or "none"
a.type	"N", "SO" or "S"
dist.type	"P", "NB" or "ZIP"
B	number of broods
cov_sim_type	how covariate data is simulated - "uniform" or "seasonal"
n.iter	number of iterations
dpw	number of days per week to simulate
meth	optimisation method - "Nelder-Mead", "SANN" or "L-BFGS-B"
convert_weekly	convert from the daily to the weekly format - "yes" or "no"
vary_phi	vary phi over the season - "yes" or "no"
NA_type	missing values - "match_NA", "prop_NA" or "none"
nS	number of sites, if matching raw data then use the length of the output
year	year to simulate

Value

List of the output for each iteration of the simulation.

Examples

```

library(ButterflyLifespan)

# setup simulation parameters
year <- "2018"
setup_level <- "daily"
env_cov.type <- "none"
a.type <- "SO"
dist.type <- "P"
B <- 1 ## single brood
cov_sim_type <- "uniform"
n.iter <- 250
dpw <- 1
meth <- "Nelder-Mead"
convert_weekly <- "no"
vary_phi <- "no"

```



```
NA_type <- "prop_NA"
nS <- 200

# run simulation
results <- simulation(setup_level,
  env_cov.type,
  a.type,
  dist.type,
  B,
  cov_sim_type,
  n.iter,
  dpw,
  meth,
  convert_weekly,
  vary_phi,
  NA_type,
  nS,
  year)
```

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